

1 Solution — GIAM 9.2

Problem 1

1.1 Problem

Do the algebra (and show all your work!) to prove that the invariant defined in this section actually has the value 1 for the set of all the men occupying the x -axis and the lower half-plane.

1.2 Setup — Reminder of the Invariant

Recall the setup from Section 9.2:

- An “army” of checkers occupies every lattice point $(x, y) \in \mathbb{Z}^2$ with $y \leq 0$.
- The target is the position $(0, 5)$ — five steps above the x -axis.
- To each lattice point p we assign a value $r^{d(p)}$, where $d(p)$ is the *taxicab distance* from p to the target. For an army point (x, y) with $y \leq 0$:

$$d(x, y) = |x| + |y - 5| = |x| + (5 - y).$$

- The number r is chosen as the positive root of $x^2 + x - 1 = 0$, so $r = \frac{\sqrt{5}-1}{2} = 1/\varphi$ and

$$r^2 + r = 1 \quad \Longleftrightarrow \quad r^2 = 1 - r.$$

- The *invariant* S is the total value of the army:

$$S = \sum_{(x,y): y \leq 0} r^{d(x,y)}.$$

Goal. Show $S = 1$.

1.3 Counting the Army by Distance Ring

The army points at fixed taxicab distance $5 + k$ from $(0, 5)$ form a “diamond half-ring” — the bottom half of a taxicab diamond centered at the target, intersected with $y \leq 0$.

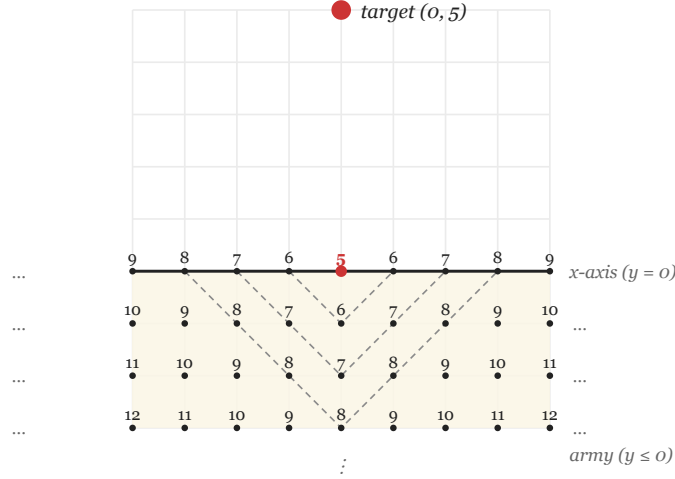


Figure 1.1: Lattice points in the army labeled with their taxicab distance from the target $(0, 5)$. Dashed V-lines join the points at distance $d = 6, 7, 8$ — visible counts 3, 5, 7 along the V’s confirm the $2k+1$ pattern.

Claim. At taxicab distance $5 + k$ from $(0, 5)$ there are exactly $2k + 1$ army points.

Proof. A point (x, y) with $y \leq 0$ has distance $5 + k$ from $(0, 5)$ iff $|x| + (5 - y) = 5 + k$, i.e. $|x| = k - j$ where $j := -y \in \{0, 1, \dots, k\}$. For each such j :

- if $j = k$: $|x| = 0$, giving one point $(0, -k)$;
- if $0 \leq j < k$: $|x| = k - j > 0$, giving two points $(\pm(k - j), -j)$.

Total: $1 + 2k = 2k + 1$. ■

The figure makes this visible at small k :

k	Distance $5 + k$	Army points at that distance	Count $2k + 1$
0	5	$(0, 0)$	1
1	6	$(\pm 1, 0), (0, -1)$	3

2	7	$(\pm 2, 0), (\pm 1, -1), (0, -2)$	5
3	8	$(\pm 3, 0), (\pm 2, -1), (\pm 1, -2), (0, -3)$	7

Table 1.1

1.4 Assembling the Sum

Group the army by distance ring:

$$S = \sum_{(x,y): y \leq 0} r^{d(x,y)} = \sum_{k=0}^{\infty} (2k+1) r^{5+k} = r^5 \sum_{k=0}^{\infty} (2k+1) r^k.$$

Define

$$T := \sum_{k=0}^{\infty} (2k+1) r^k = 1 + 3r + 5r^2 + 7r^3 + 9r^4 + \dots$$

So $S = r^5 T$. The work is now to compute T .

1.5 Computing T — Arithmetic-Geometric Trick

The series for T is *arithmetic-geometric*: coefficients $2k+1$ grow linearly while powers r^k shrink geometrically. The standard manoeuvre is to multiply by r and subtract:

$$\begin{aligned} T &= 1 + 3r + 5r^2 + 7r^3 + 9r^4 + \dots \\ rT &= r + 3r^2 + 5r^3 + 7r^4 + \dots \end{aligned}$$

Subtract row 2 from row 1:

$$\begin{aligned} T - rT &= 1 + (3-1)r + (5-3)r^2 + (7-5)r^3 + (9-7)r^4 + \dots \\ &= 1 + 2r + 2r^2 + 2r^3 + 2r^4 + \dots \\ &= 1 + 2(r + r^2 + r^3 + \dots). \end{aligned}$$

The tail is a geometric series with ratio r , convergent since $|r| < 1$:

$$r + r^2 + r^3 + \dots = \frac{r}{1-r}.$$

Therefore

$$T - rT = 1 + \frac{2r}{1-r} = \frac{(1-r) + 2r}{1-r} = \frac{1+r}{1-r}.$$

Since $T - rT = (1-r)T$:

$$(1-r)T = \frac{1+r}{1-r} \implies T = \frac{1+r}{(1-r)^2}.$$

1.6 Putting It Together

Substituting back into $S = r^5 T$:

$$S = r^5 \cdot \frac{1+r}{(1-r)^2} = \frac{r^5(1+r)}{(1-r)^2}.$$

1.7 Simplifying — One Pass of $r^2 = 1 - r$

The defining identity $r^2 + r = 1$ rearranges to $1 - r = r^2$. So

$$(1-r)^2 = (r^2)^2 = r^4.$$

Substituting into the denominator:

$$S = \frac{r^5(1+r)}{r^4} = r(1+r) = r + r^2.$$

And $r + r^2 = 1$ — the defining identity again. Hence

$$S = 1. \quad \blacksquare$$

1.8 Remark — A Single Identity Does All the Work

The whole simplification rests on *one* identity:

$$r^2 + r = 1.$$

It is invoked twice — once as $1 - r = r^2$ to collapse the denominator $(1 - r)^2$ to r^4 , and once as $r + r^2 = 1$ to finish off the numerator. A single quadratic relation suffices to evaluate the entire two-dimensional sum.

This is precisely why $r = 1/\varphi$ is the *right* value. The same identity was *engineered* in Section 9.2 so that every legal checker-jump satisfies $r^{k+2} + r^{k+1} \leq r^k$ — i.e. so a jump never *increases* S . The two properties of r — jump-monotonicity and a clean total mass — are not independent coincidences; they are both consequences of $r^2 + r = 1$.

1.9 Remark — Alternative Derivation: Separate the x - and y -Sums

Since the distance splits as $d(x, y) = |x| + (5 - y)$, the exponential $r^{d(x, y)} = r^{|x|} \cdot r^{5-y}$ *factors*, and so does the sum:

$$S = \sum_{y \leq 0} \sum_{x \in \mathbb{Z}} r^{|x|} \cdot r^{5-y} = \left(\sum_{x \in \mathbb{Z}} r^{|x|} \right) \cdot \left(\sum_{y \leq 0} r^{5-y} \right).$$

Both pieces are *purely geometric* (no arithmetic-geometric trick needed):

$$\sum_{x \in \mathbb{Z}} r^{|x|} = 1 + 2 \sum_{x=1}^{\infty} r^x = 1 + \frac{2r}{1-r} = \frac{1+r}{1-r},$$

$$\sum_{y \leq 0} r^{5-y} = \sum_{j=0}^{\infty} r^{5+j} = \frac{r^5}{1-r} \quad (\text{setting } j = -y).$$

Multiplying:

$$S = \frac{1+r}{1-r} \cdot \frac{r^5}{1-r} = \frac{r^5(1+r)}{(1-r)^2}.$$

Same answer, *without* invoking the arithmetic-geometric series. The diamond-ring counting (main proof) is more *geometric* — it makes the 1, 3, 5, 7, ... pattern visible — but the factored form is *algebraically* leaner.

1.10 Remark — Why “Five Steps” is the Punchline

The fact $S = 1$ is the entire proof of the five-step impossibility, packaged in a single number. Here is how the punch lands:

- A man at the target $(0, 5)$ contributes $r^0 = 1$ — by itself, *all* of S .
- Every checker-jumping move *preserves or decreases* S . (Selectability of r as a root of $r^2 + r - 1 = 0$ was forced by this requirement.)
- The army starts with total value 1 .

So *after any finite sequence of jumps*, the total value of the configuration is ≤ 1 . To place a man at $(0, 5)$, the *post-move* configuration must include that single position worth 1 — and the remaining men (still on finite-distance lattice points) contribute *strictly positive* extra value. The total would then exceed 1 , contradicting monotonicity.

Hence the target $(0, 5)$ is unreachable in finitely many jumps. The whole proof is just $S = 1$ together with the monotonicity inequality — and *both* come from the same root of $x^2 + x - 1$.

1.11 Remark — Convergence is Not Free

The sum S ranges over a countable two-dimensional lattice. Convergence isn't automatic — it works because $r < 1$ (in fact $r = 1/\varphi \approx 0.618$), which forces both the x - and y -marginals to decay geometrically.

Concretely, each row $y = -j$ in the army contributes

$$r^{5+j} \cdot \sum_{x \in \mathbb{Z}} r^{|x|} = r^{5+j} \cdot \frac{1+r}{1-r},$$

a single geometric series in j with ratio r .

If we had picked $r \geq 1$, the sum would diverge and the entire “invariant” approach would collapse — there would simply be no finite number to compare against. The choice $r = 1/\varphi$ is in this sense doubly tight: it is the *largest* value of r for which (i) the jump-monotonicity inequality $r^{k+2} + r^{k+1} \leq r^k$ holds, and (ii) $|r| < 1$ so the sum converges. Any smaller r also works for (i) and (ii), but yields $S < 1$ and is too weak to prove the five-step bound.

1.12 Remark — Why the Numerator Doesn't Collapse to a Power of r

The simplification $(1 - r)^2 = r^4$ uses the identity *once*. The numerator $r^5(1 + r) = r^5 + r^6$ does not collapse to a single power of r — but we don't need it to. Using $r^2 = 1 - r$:

$$r^5 + r^6 = r^4(r + r^2) = r^4 \cdot 1 = r^4.$$

So $S = r^5(1 + r)/(1 - r)^2 = r^4/r^4 = 1$. This is an equally clean way to land: factor r^4 out of the numerator and pair it with $r + r^2$, which the identity hands you as 1.

In summary the algebra has *two* good simplifying strategies, both keyed on $r^2 + r = 1$:

1. Reduce $(1 - r)^2$ to r^4 (denominator), then use $r + r^2 = 1$ (numerator).
2. Expand the numerator $r^5 + r^6$, then use $r + r^2 = 1$ inside the factored form $r^4(r + r^2)$.

Both arrive at $S = 1$ in one or two pen-strokes.